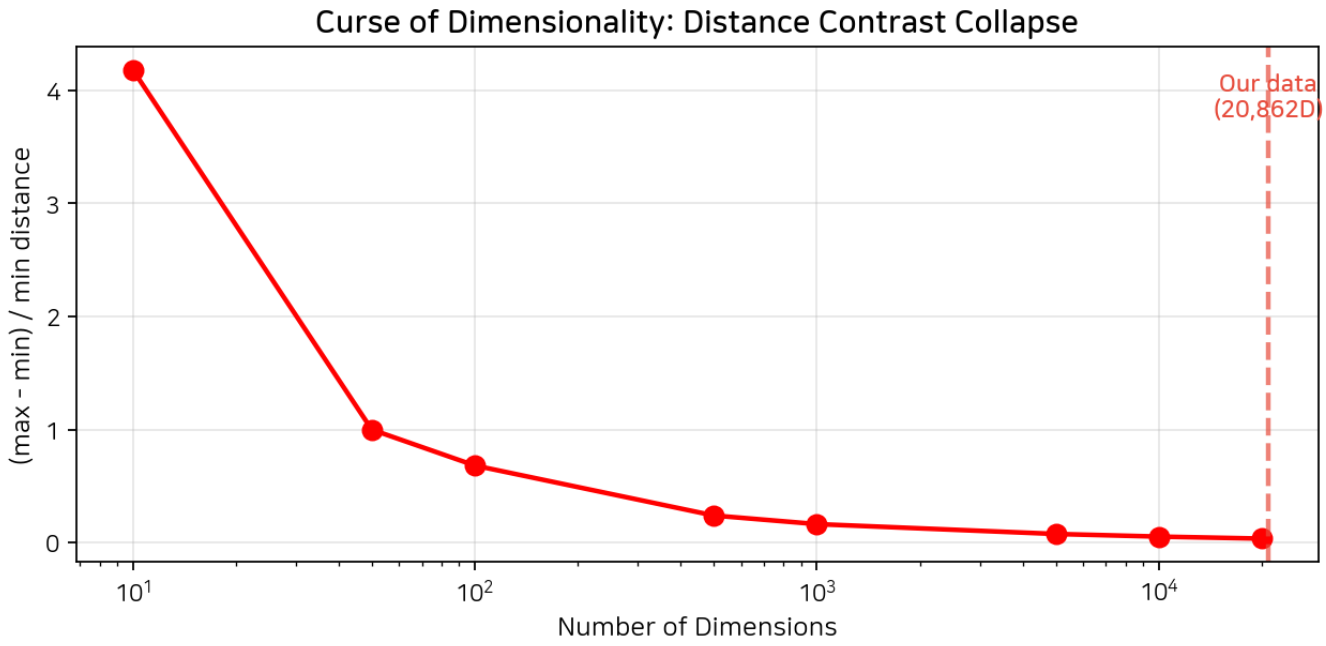
Mathematical Principles & Theorems

Topological Autoencoder with COAST Loss

Complete mathematical reference for all algorithms in the pipeline

0. Motivation: Curse of Dimensionality



Theorem (Beyer et al., 1999) — Distance Concentration

For i.i.d. random vectors in d dimensions, as d → ∞:

```
lim (d→∞) (max(D) - min(D)) / min(D) → 0
```

All pairwise distances converge to the same value. Euclidean distance loses its discriminative power.

Implication: With 20,862 genes, Euclidean pairwise distances between patients collapse. The topological loss based on Euclidean distances receives weak gradient signals, making it difficult to preserve neighborhood structure in the latent space. This motivates the use of angular metrics (cosine, Pearson) which are more robust in high dimensions.

1. Log Transformation

Global log1p transformation:

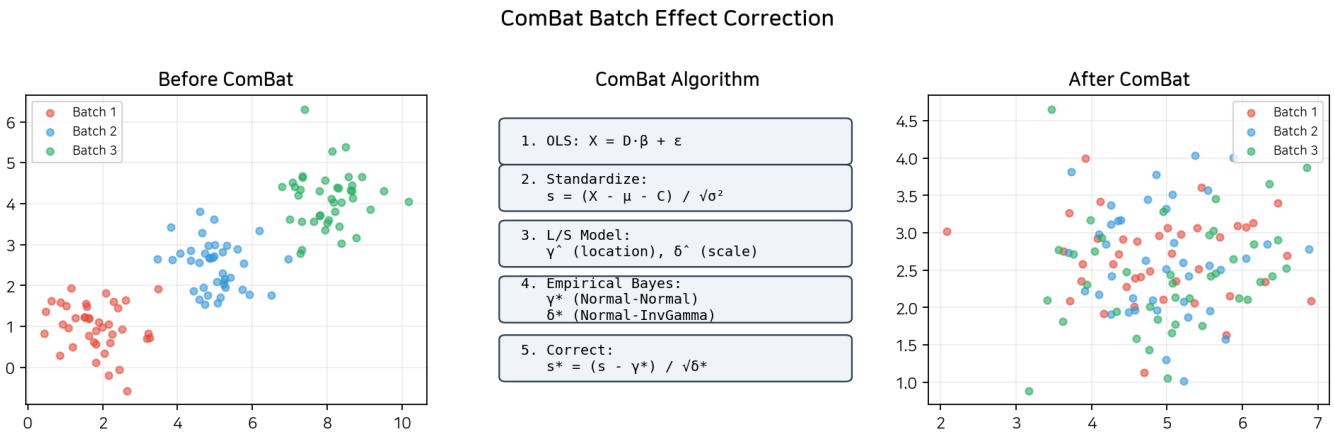
```
x' = log(1 + x)
```

Property	Description
Domain	$x \geq 0$ (TPM values are non-negative)
Zero preservation	$\log(1 + 0) = 0$
Variance stabilization	Compresses range $[0, 10^6]$ to $[0, \sim 14]$
Distribution symmetry	Right-skewed TPM → approximately symmetric

Why global (not selective)? The pipeline uses cosine distance, which measures angular divergence. Selectively applying log1p (only to genes exceeding skewness/kurtosis thresholds) creates gene-dependent nonlinear distortions that break the directional assumptions cosine relies on. Global log1p compresses all genes consistently, preserving relative angular structure.

2. ComBat Batch Effect Correction

Johnson, Li, & Rabinovic (2007). Removes systematic technical variation across 31 tissue source sites while preserving biological signal.



Step 1: Design Matrix & OLS

```
D = [B | C] (n_samples × n_design)
B = one-hot batch indicators
C = categorical/continuous covariates (Target: tumor/normal)

OLS: β̂ = (D^T D)^-1 D^T X^T
```

Step 2: Standardization

```
Grand mean (batch-size weighted):
  μ̂_j = Σ_i (n_i / N) · β̂_ij

Pooled variance:
  σ̂²_pooled,j = (1/N) · Σ (X_j - D·β̂_j)²

Standardize:
  s_j = (X_j - μ̂_j - Ĉ_j) / √σ̂²_pooled,j
```

Step 3: Location-Scale (L/S) Model

```
For batch i, gene j:
  γ̂_ij = location shift (OLS on standardized data)
  δ̂_ij = var(s_j in batch i)   (scale factor)
```

Step 4: Empirical Bayes Shrinkage

Conjugate Priors

Location (Normal-Normal):

Prior: $\gamma_j \sim N(\bar{\gamma}_j, \tau^2_j)$

Posterior mean:

$$\gamma^*_{ij} = (\tau^2_j \cdot n_i \cdot \hat{\gamma}_{ij} + \delta_{old,j} \cdot \bar{\gamma}_j) / (\tau^2_j \cdot n_i + \delta_{old,j})$$

Scale (Normal-Inverse Gamma):

Hyperparameters: $a_{prior} = (2s^2 + m^2) / s^2$, $b_{prior} = (ms^2 + m^3) / s^2$

Posterior mean:

$$\delta^*_{ij} = (0.5 \cdot \Sigma(s - \gamma^*)^2 + b_{prior}) / (n_i/2 + a_{prior} - 1)$$

Iterate until convergence (< 0.0001).

Step 5: Final Correction

$$s_{corrected} = (s - \gamma^*) / \sqrt{\delta^*}$$

Restore scale:

$$X_{corrected} = s_{corrected} \cdot \sqrt{\sigma^2_{pooled}} + \hat{\mu} + \hat{C}$$

3. Topological Loss Function

Total Loss:

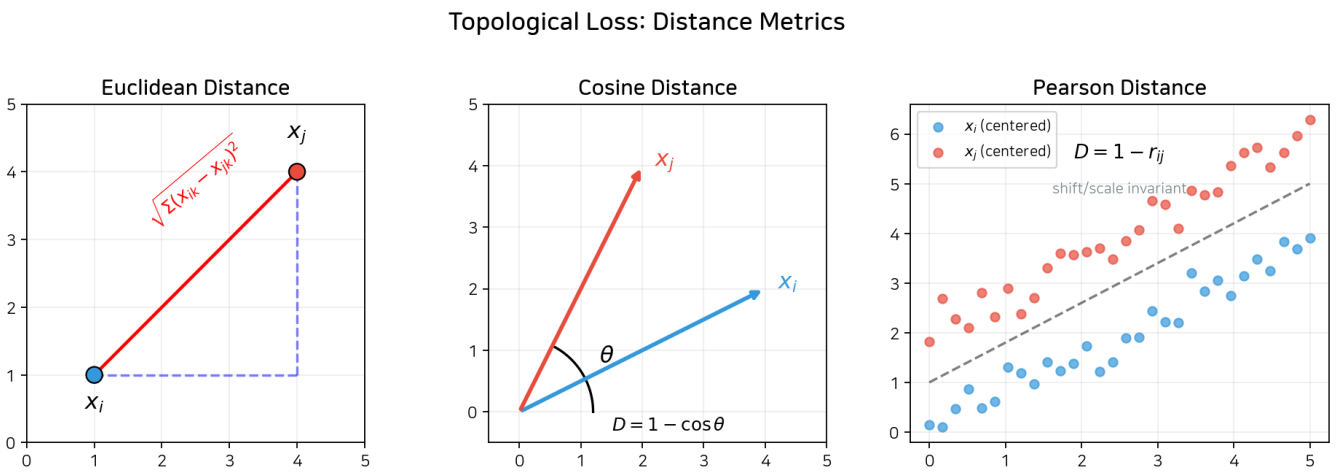
$$L_{total} = L_{recon} + \lambda \cdot L_{topo}$$

$$L_{recon} = (1/n) \Sigma_i ||x_i - \hat{x}_i||^2 \quad \text{(MSE)}$$

$$L_{topo} = MSE(\tilde{D}_{latent}, \tilde{D}_{original})$$

where $\tilde{D} = D / \max(D)$ (normalized to [0, 1])

3.1 Distance Metrics



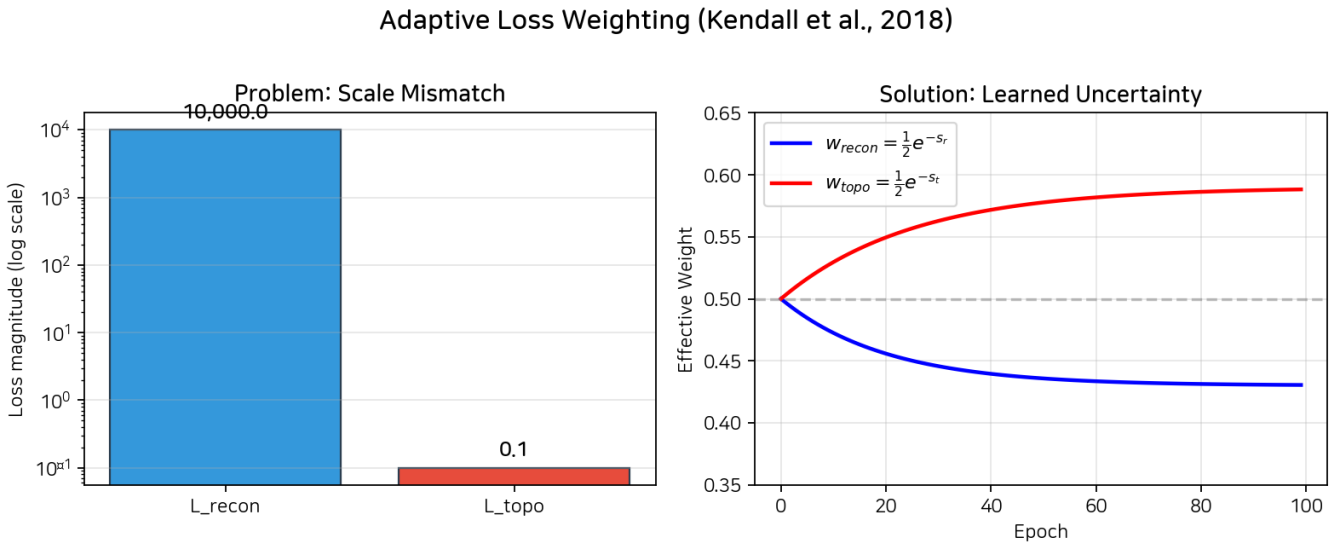
Metric	Formula	Range	Invariant to
Euclidean	$D_{ij} = \sqrt{(\Sigma_k (x_{ik} - x_{jk})^2 + \varepsilon)}$	$[0, \infty)$	Nothing
Cosine	$D_{ij} = 1 - (x_i \cdot x_j) / (\|\|x_i\|\| \cdot \|\|x_j\|\|)$	[0, 2]	Magnitude (scaling)
Pearson	$D_{ij} = 1 - r_{ij}$ (= cosine of mean-centered vectors)	[0, 2]	Scaling + shift

Why Cosine works better in high dimensions

Cosine similarity is computed as a normalized inner product, producing values in [-1, 1] regardless of dimensionality. Unlike Euclidean distance, the signal-to-noise ratio of cosine does not degrade as dimensions increase, because normalization by vector magnitude prevents noise accumulation across dimensions.

4. Adaptive Loss Weighting

Kendall, Gal, & Cipolla (2018). "Multi-Task Learning Using Uncertainty to Weigh Losses."



Homoscedastic Uncertainty Weighting

For two tasks with losses L_1, L_2 and learnable log-variances $s_1 = \log(\sigma_1^2)$, $s_2 = \log(\sigma_2^2)$:

$$L = \frac{1}{2} e^{-s_1} L_1 + \frac{1}{2} e^{-s_2} L_2 + \frac{1}{2} s_1 + \frac{1}{2} s_2$$

Derivation: From maximizing the Gaussian log-likelihood of each task:

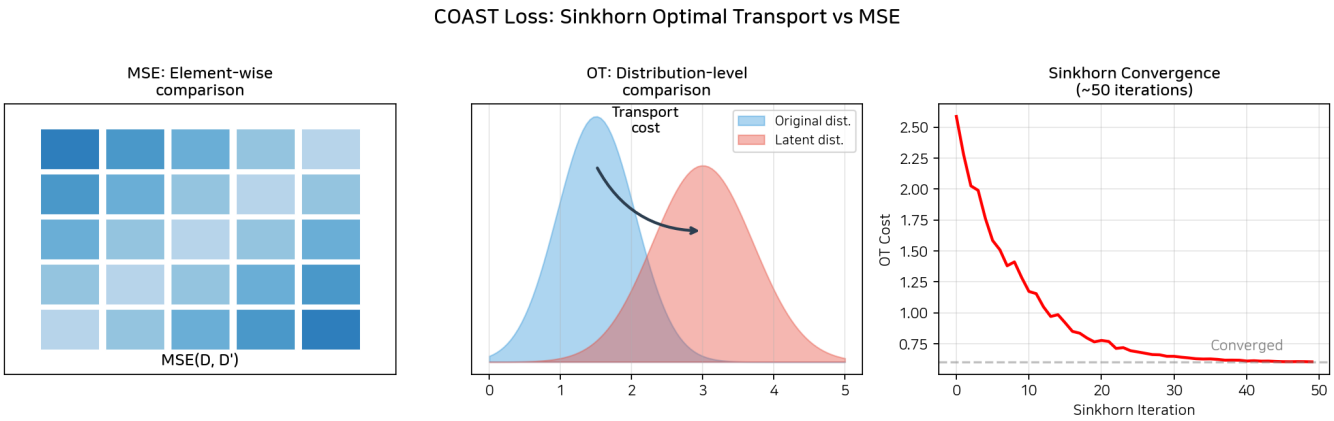
$\log p(y \mid f(x)) = -\frac{1}{2} \|y - f(x)\|^2 / \sigma^2 - \frac{1}{2} \log \sigma^2 + \text{const}$

Properties:

- $\sigma \to \infty$ would zero out the loss, but $\log(\sigma)$ regularizer prevents this
- At convergence, the weight ratio reflects intrinsic task difficulty
- s_1, s_2 are trained jointly with model parameters via Adam

5. COAST Loss — Sinkhorn Optimal Transport

COAST = Cosine Optimized Adaptive Sinkhorn Transport.
Based on Cuturi (2013). "Sinkhorn Distances: Lightspeed Computation of Optimal Transport."



5.1 Distance to Probability Conversion

Gaussian RBF Kernel:

$$S_{ij} = \exp(-D_{ij}^2 / 2\sigma^2)$$

Row-normalize to probability:

$$P_{ij} = S_{ij} / \Sigma_k S_{ik}$$

```
σ is a learnable parameter (initialized to median pairwise distance).
Diagonal is masked (self-distances excluded).
```

5.2 Entropy-Regularized Optimal Transport

Kantorovich Problem with Entropic Regularization

$$OT_\varepsilon(a, b) = \min_{\pi \in \Pi(a,b)} \langle C, \pi \rangle + \varepsilon \cdot KL(\pi \,||\, a \otimes b)$$

where:

- $a, b \in \Delta^n$ are probability distributions (from Step 5.1)
- C is the cost matrix (pairwise distances in latent space)
- π is the transport plan
- $\Pi(a,b) = \{ \pi \geq 0 : \pi 1 = a, \pi^T 1 = b \}$
- $\varepsilon > 0$ is the regularization strength (smaller = more accurate, less stable)

5.3 Sinkhorn Algorithm (Log-Domain)

Pre-compute:

```
log_C = -C / ε
```

Iterate (50 times):

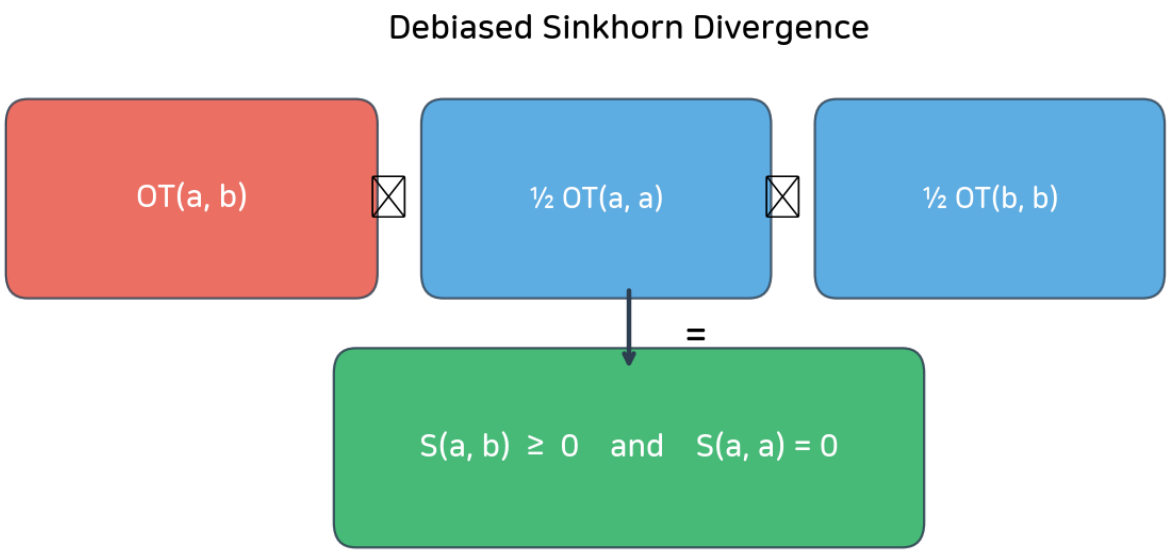
```
g ← ε · log(b) - ε · logsumexp(f + log_C, dim=rows)
f ← ε · log(a) - ε · logsumexp(log_C + g, dim=cols)
```

Final cost:

```
OT = ⟨a, f⟩ + ⟨b, g⟩
```

Why log-domain? Standard Sinkhorn uses exp(-C/ε), which underflows to 0 when ε is small. Log-domain computation uses logsumexp, which is numerically stable for any ε. Complexity: $O(n^2)$ per iteration, vs $O(n^3)$ for exact Wasserstein (Hungarian algorithm).

5.4 Debiased Sinkhorn Divergence



Debiased Divergence (Feydy et al., 2019)

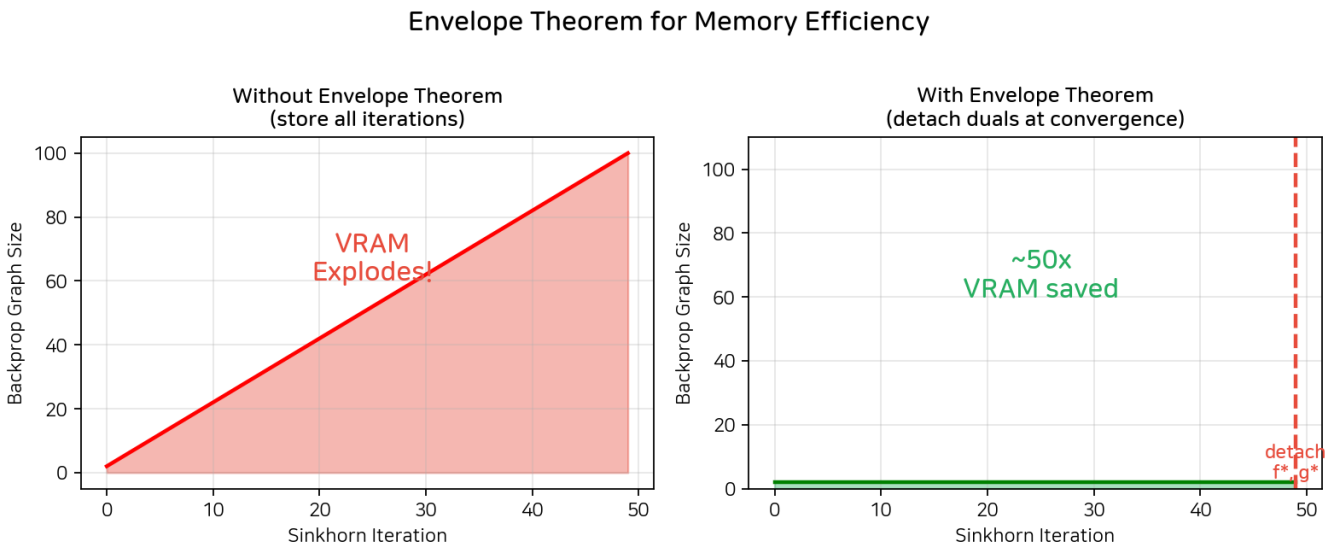
Raw Sinkhorn OT cost can be negative (not a true divergence). The debiased version:

$$S(a, b) = OT_\varepsilon(a, b) - \tfrac{1}{2} OT_\varepsilon(a, a) - \tfrac{1}{2} OT_\varepsilon(b, b)$$

Guarantees:

- $S(a, b) \geq 0$ (non-negative)
- $S(a, a) = 0$ (identity of indiscernibles)
- Metrizes weak convergence (topologically meaningful)

5.5 Envelope Theorem for Backpropagation



Envelope Theorem (Danskin's Theorem)

At the optimal dual variables f*, g*:

```
∂OT / ∂θ = ∂/∂θ [⟨a, f*⟩ + ⟨b, g*⟩]
```

The gradient of the optimal value function equals the gradient of the objective evaluated at the optimum, treating f*, g* as constants.

Implementation: After Sinkhorn converges, `detach()` f and g, then compute cost. Backprop only flows through a, b (the input distributions), not through the 50-iteration Sinkhorn loop. Saves ~50× VRAM.

5.6 Topo Multiplier

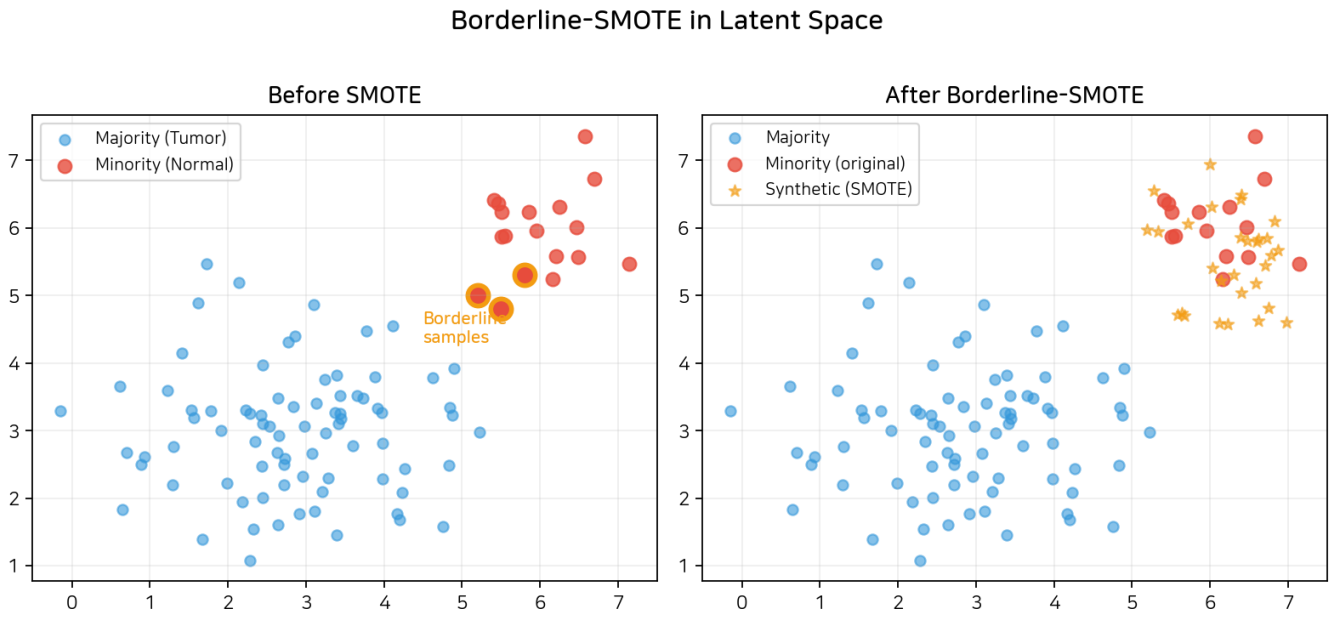
Final COAST topological loss with multiplier m:

$$L_{topo}^{COAST} = (1/n)\ \Sigma_i\ m \cdot S(P_i^{orig},\ P_i^{latent})$$

Why needed? Sinkhorn distances are inherently small (~0.6) while reconstruction MSE is ~10,000. Even with adaptive weighting, the topo gradients are negligible without scaling. The multiplier m (tuned via log-scale sweep, sweet spot ~1,000–50,000) brings topo gradients to a magnitude where the adaptive weighting can effectively balance both objectives.

6. Borderline-SMOTE

Han, Wang, & Mao (2005). Applied in latent space for class balance augmentation.



Algorithm:

```
For each minority sample x_i:
  1. Find k nearest neighbors (all classes)
  2. If > k/2 neighbors are majority class ⇒ x_i is "borderline"
  3. For borderline x_i, pick random minority neighbor x_nn:

    x_new = x_i + λ · (x_nn - x_i),     λ ~ Uniform(0, 1)
```

Why in latent space (not original 20,862D)?

- k-NN is more meaningful in low dimensions (curse of dimensionality avoided)
- TAE's topological loss ensures latent space preserves structural relationships
- Computationally much faster

WARNING: Never decode SMOTE outputs. Linear interpolation in latent space does not respect the nonlinear encoder manifold. Decoded points would be biologically meaningless. Use SMOTE-augmented latent vectors only for classifier training.

7. Statistical Tests (Exploratory Analysis)

Test	Formula	Purpose
Welch's t-test	$t = (\mu_1 - \mu_2) / \sqrt{(s_1^2/n_1 + s_2^2/n_2)}$	Differential expression (unequal variance)
Point-Biserial r	$r_{pb} = (M_1 - M_0)/s_n \cdot \sqrt{(n_0 n_1/n^2)}$	Gene-class correlation (continuous × binary)
Chi-Square	$\chi^2 = \Sigma(O_{ij} - E_{ij})^2 / E_{ij}$	TSS × tumor independence
Cramer's V	$V = \sqrt{(\chi^2 / (N \cdot \min(r-1, c-1)))}$	Effect size for categorical association
Mann-Whitney U	Non-parametric rank-sum test	Bad vs. good gene |PB_Corr| distributions

Volcano Plot Criteria

```
x-axis: Log2 Fold Change = log2(μtumor / μnormal)
y-axis: -log10(p-value)

Significance thresholds: |Log2FC| > 1 and p < 0.05
```

8. Summary of Mathematical Components

Stage	Method	Key Principle	Complexity
Preprocessing	log1p	Variance stabilization, angular preservation	$O(n \cdot d)$
Preprocessing	ComBat	Empirical Bayes shrinkage (conjugate priors)	$O(n \cdot d \cdot b)$
Architecture	Autoencoder	Nonlinear dimensionality reduction	$O(n \cdot d \cdot h)$
Topo Loss	Distance Matrix MSE	Pairwise distance preservation	$O(n^2 \cdot d)$
Topo Loss	Cosine/Pearson	Angular robustness in high-D	$O(n^2 \cdot d)$
Weighting	Adaptive (Kendall)	Homoscedastic uncertainty	$O(1)$ extra params
Topo Loss	**COAST**	**Cosine + Adaptive + Sinkhorn OT**	$\mathbf{O(n^2 \cdot T)}$
COAST	Debiased divergence	Non-negative, metrizes weak convergence	3× OT cost
COAST	Envelope theorem	Danskin's theorem (detach duals)	~50× VRAM saved
Augmentation	Borderline-SMOTE	Boundary-focused oversampling	$O(n \cdot k \cdot d_{lat})$
Evaluation	Logistic Regression	Linear separability test	$O(n \cdot d_{lat})$